

- 2 (a) Fig. 2.1 shows a cell at one of the main stages of mitosis in the mitotic cell cycle.

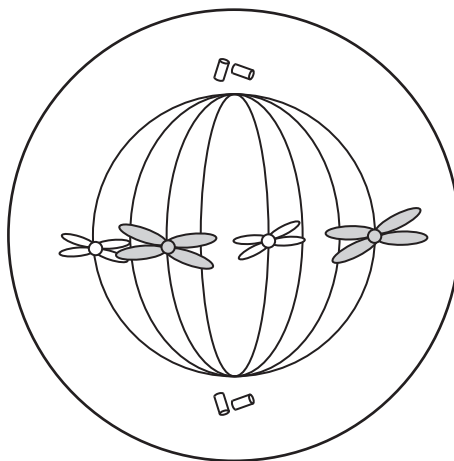


Fig. 2.1

- (i) Name the stage of mitosis shown in Fig. 2.1.

..... [1]

- (ii) Fig. 2.2 shows the cell in Fig. 2.1 at the start of cytokinesis.

Complete Fig. 2.2 to show the daughter chromosomes in each nucleus.

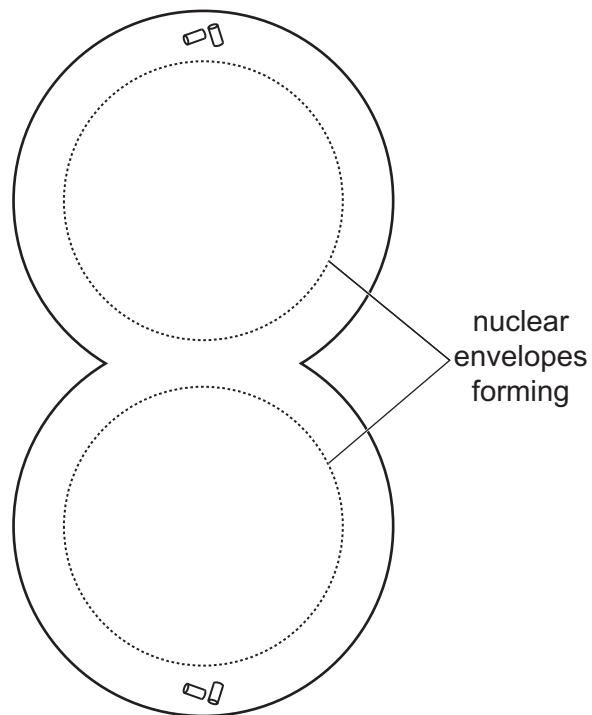


Fig. 2.2

[2]

(b) State the role of telomeres during DNA replication.

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..... [1]

(c) Multiple myeloma is a type of cancer in the bone marrow where some of the stem cells start to produce abnormal blood cells.

- One treatment is to collect stem cells from the bone marrow of the person with multiple myeloma. Healthy stem cells are isolated and grown in the laboratory.
- Radiation is then used to destroy all stem cells and cancerous cells in the bone marrow.
- Finally, large numbers of the healthy stem cells grown in the laboratory are returned to the bone marrow.

Suggest the role of stem cells in this treatment of multiple myeloma.

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..... [3]

[Total: 7]